

United State's EV Infrastructure Evaluation: Now and Future

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Background

The International Energy Agency (IEA) reported that global Electric Vehicle (EV) market “share **increased from 4% in 2020 to 18% in 2023**” [1]. This upward trend demonstrates the increasing consumer demand for electric vehicles and a more sustainable livelihood; something many of us have surely noticed in our daily lives.

With the increase of demand for EV's, there will be a commensurate increase in demand for supporting infrastructure. However, the Trump administration announced, **on February 6, 2025, the ending the National Electric Vehicle Infrastructure (NEVI) Formula Program - a program to support the expansion of EV charging ports to 500,000 by 2030** [2].

President Trump's concept of a plan is to allow private companies like Tesla, headed by the current head of the U.S. Department of Government Efficiency, Elon Musk, to dictate the future of EV infrastructure in the U.S.

While markets can be efficient at unifying supply with demand at a profitable price point, they inevitably exclude those who face financial barriers. **This can lead to unmet demand in one region and oversupply in another, reinforcing existing socioeconomic disparities.**

Anticipating this shift in policy, we identified key considerations in the current EV infrastructure development statewide where improvements ensure sustainability and equity in our EV infrastructure policy.

Ethical Considerations and Motivations

Easy access to charging infrastructure plays a paramount role in a consumers' decisions when buying electric vehicles. This project evaluates available data to determine who is being served by the current infrastructure investment and who faces the largest barriers to entry in the current marketplace.

Regional differences, such as grid capacity, energy sources, and population density involving

the existing infrastructure, should be investigated to provide data driven insights on important decisions about the future transportation infrastructure.

To ensure these investments are effective and equitable, our team has identified key questions that we believe are essential for policymakers to consider:

Questions

1. Are there currently enough charging stations within the U.S. to charge all electric vehicles?
2. If there is a deficit, how many additional EV chargers per state are needed?
3. How much money is needed to increase the infrastructure to meet the EV demand?
4. Where should infrastructure improvements prioritize not only increasing EV charging availability but also modernizing the energy grid to maximize the benefits of EV adoption?
5. Do some states see more benefits from EV adoption than others due to non-renewable resources?

Methodology : Data Collection and Sources

Data Set	Short Description	Format	Size	Time Frame	Location / Links
Alternate Fuel Stations	A data set used to power a Alternative Fuel station locator tool for the US and Canada. Filters applied to get data download are Country=US and Canada and State/Province/Territory= "All"	CSV	93,826	Through Jan 2025	Afdc.energy.gov
EV Ownership Counts	A summary table from the US Department of Energy's website. It contains the number of light duty vehicles for each state by fuel type	HTML Table	51	Through 2023	afdc.energy.gov
National Household Travel Survey	National Household Travel Survey data is available as a zip file for download that contains multiple CSV files. Contains various information regarding driving habits such as annual miles driven. Only the csv file which is specific to a vehicle was used from the set.	CSV	14,684	Survey 2022	nhts.ornl.gov
Power Generation Emissions	Emissions and Generation Resources Integrated Database. Data set that contains the types of power generation methods used for each state along with the emissions generated.	XLSX	52	Released Jan 15 2025	epa.gov
State names, abbreviations, and Location table	This is a helper table to go between State names, central geographical locations (latitude and longitude) and 2 letter abbreviations where required.	HTML Table	51	N/A	BU.edu

Methodology : Data Manipulation

The objective is to have a Pandas dataframe that contain the following information:

- **State**
- **Number of EV's**
- **EV Charging Port Quantity**
- **Average Miles Driven**
- **Electrical Capacity Needs (kWh)**
- **Electrical Capacity Available (kWh)**
- **Carbon Dioxide Generation (grams of Co2 per kwh)**

With this information, we will generate the required visualizations and perform the required analysis.

Google Colab Usage

Given Google Colab was utilized to create visualizations and clean data, a method for anyone to acquire the data sources is needed to run the code. The raw files are stored on Google drive, shared with “everyone”, and the file URL is included in a code. **A function was created for reading each file type using the file ID, which included the URL .**

EV Charging Ports

This information came from a csv data set for the alternate fuel station locations within the US and Canada. This csv file contained more columns than were applicable to our project, so the relevant columns were loaded into the dataframe. The `pandas.read_csv` function was used to accomplish this using `parameters` `usecols` and `dtype`. All non-US charging station information was filtered out (`['Country']=='US'`) and only public charge stations were kept: (`['EV Workplace Charging']=='False'`).

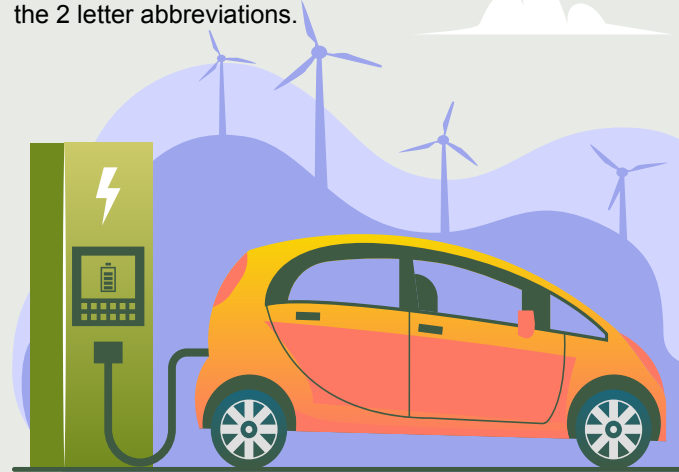
A helper table was created and used acquire both states full names and the 2 letter abbreviations. This table was taken from a web page (HTML Table) and loaded into a dataframe. Some columns were irrelevant and dropped, while other columns were renamed to match with other datasets.

To summarize the remaining 70,000+ rows, the total quantity of chargers for each charger level was aggregated by state using the `groupby()` method.

EV Population

The EV population comes from an HTML table that is available on a web page for the US Department of Energy. The `pandas.read_html()` function was used to accomplish this. The [Beautiful Soup](#) library was used to help with pulling the table data from the HTML code.

In addition, the `String IO` module was also used to feed the html into the `pandas.read_html()` function and create a new dataframe. This dataset was already a summary table by State, but had to be merged with the helper table to get the 2 letter abbreviations.



Methodology : Data Manipulation and Consolidation

EV Mileage

The EV mileage data derives from the National Housing and Travel survey. This information was acquired from a csv file using the `pandas.read_csv` function. This data reflected survey data for all types of data and was filtered to reflect Battery EV's (`['VEHFUEL']=='5'`).

Unfortunately, the mileage dataset contained regional information while all other datasets were at the state level. Since it was difficult to find a source at the state level, all states included in that region were given the same average miles driven in a year. A data frame was created to associate regions to States and then merged with the EV annual mileage data.

Grid Emissions

The grid emissions data was imported using the `pandas.read_excel()` function. To incorporate state abbreviations, the `.map` method was utilized. States with NaN values for emissions were removed from the dataframe. A calculated column, called **Renewable Percentage**, provided a percentage of Renewable power of the total power produced.

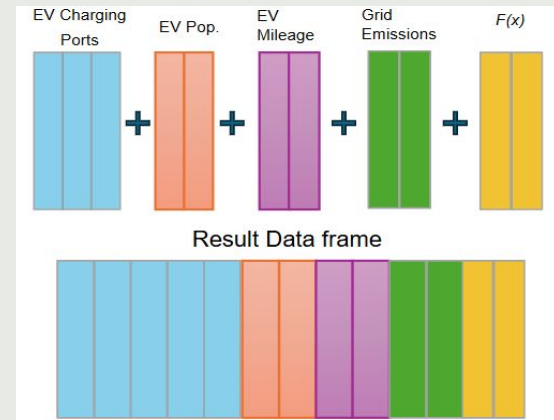
Calculated Columns

To calculate the theoretical available charge capacity at public charge stations, **18 hours per day** was used for the duration of time that EV owners can charge their cars based on a study completed by National Renewable Energy Laboratory [3]. Out of the total time, additional columns were created to calculate charging rates for **level 1 at 1 kWh per hour, Level 2 at 3.79 kWh per hour and Level 3 at 39 kWh per hour according to the Alternative Fuels Data Center (AFDC)** [4]. The total public charge capacity per day for each state was calculated summing up total charge times incorporating each charge port and type at every station.

In addition, the electrical charging capacity requirements were calculated based on the number of EV's and the average annual miles driven for each State. The EV battery usage per mile was **an average of 34.6 kwh for every 100 miles driven** [5]. **"80 percent of EV charging happens at home"** for EV owners" [6]. The last calculated column took 20% of the total needs of the electrical capacity for each state to provide a more realistic demand.

Result Data frame

The result data frame is a merge of the EV Charging Ports, EV Population, EV mileage, and Grid Emissions datasets, as well as additional calculated columns. The merge was completed using the `merge()` function. The data frames were joined on the **'state_abbr'** column using the outer join parameter.



Exploratory Data Analysis: EV Infrastructure Distribution

When exploring EV infrastructure dataset, a histogram was utilized to understand the spread of both charging stations and charging ports per state. **The histogram confirmed two things:**

- **Right skewed distribution**
- **One outlier (California) that is approximately 7 standard deviations above the mean compared to the rest of the data**

To determine the relationship between EV Infrastructure and EV population, a normal distribution is best to accurately gauge the strength of the relationship between the two variables. Given the data is skewed, 3 different normalization techniques were explored:

- **Z-score normalization**
- **Min-max Values**
- **Logarithmic Values**

Results

The z-score method and Min-max scaling still reflected a right skewed distribution and an extremely high outlier. However, the Log transformation resembles a normal distribution and the outlier is.

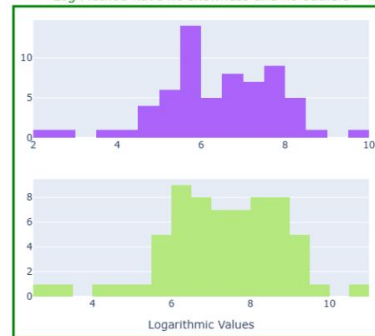
Log values of both charging stations, ports, and EV population will be utilized instead of actual values.

Normalization Technique Comparison

All techniques show right skewness and very large range with outliers



Log Method have no skewness and no outliers



Exploratory Data Analysis: EV Infrastructure versus EV Population

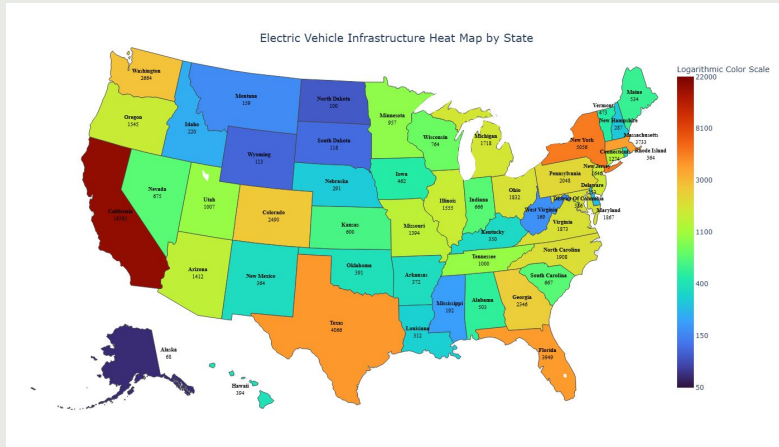
Geospatial Analysis

Geospatial heat maps are tools that help visualize and compare data related to geographical divisions. They reveal patterns based on region or nature that can be irrelevant with markers and lines.

Geospatial heat maps were used to display charging stations quantities (left map) and EV population (right map) per state. The color scales below show warmer colors for large data values and cooler colors for small data values. **Due to the normality of the log values, the color scale for both maps are scaled logarithmically.**

Results

There is an average of 1490 stations and 69,724 vehicles per state. Based on the maps, the north Mountain West states plus Alaska have low amount of EV's and charging stations. This could be due to low populations in each state. The states with the highest population, New York, Texas, California, and Florida, have high quantities of EV's and charging stations. The two maps highlight an important characteristic - **both maps exhibit similar heat levels, inferring there is a relationship between EV's and electric vehicle stations.**



Correlation Analysis: EV Population versus Charging Ports

To understand the strength of the relationship between EV's and the charging infrastructure, a scatter plot is utilized, each marker representing a state. Instead of EV charging stations, charging ports were used because of the 1:1 ratio between charging ports and EVs.

The scatter plot of ports per state and EV's per state show a positive relationship, but the trend is not linear. In addition, there is a huge outlier, the same outlier from the histogram, that is noticeable from this chart. Given there are only 51 data points, this specific point was not removed to assure a robust analysis.

Given all visualizations and analyses previous to the correlation analyses utilized log values of each variable, the same was applied to the this analyses. The log-log plot shows a positive and linear correlations, with **a correlation of fit being 0.9268.**

The linear regression of the log-log plot provides a formula that inputs a log value of EV's and returns a predicted log value of ports.

Linear Regression of Log-Log Plot:

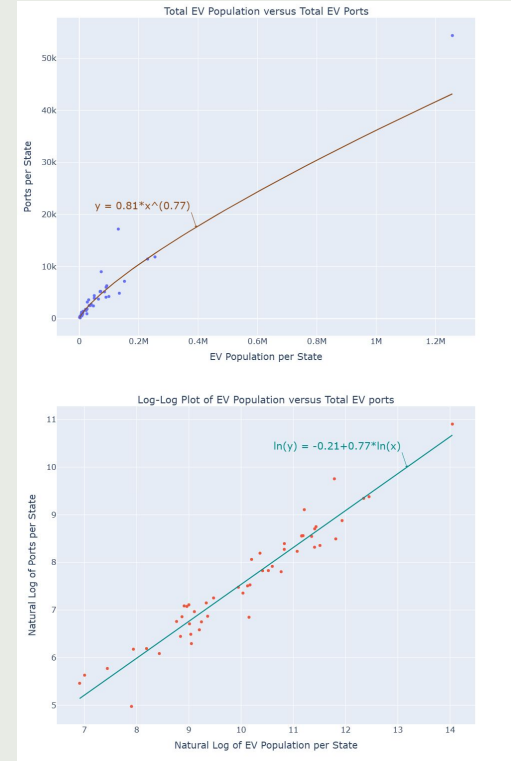
$$\ln(\text{Charging Ports}) = 0.21 + 0.77 \times \ln(\text{EV's})$$

The linear regression can be transforms into a power function, which allows for the use of actual values.

Power Function Derived from Log-Log Plot:

$$\text{Charging Ports} = 0.81 \times \text{EV's}^{0.77}$$

This power formula will be utilized to estimate how many extra ports we need to charge the forecasted number of EV's in 2030.



Predictive Analysis: Estimated 2030 EV Charging Port Needs

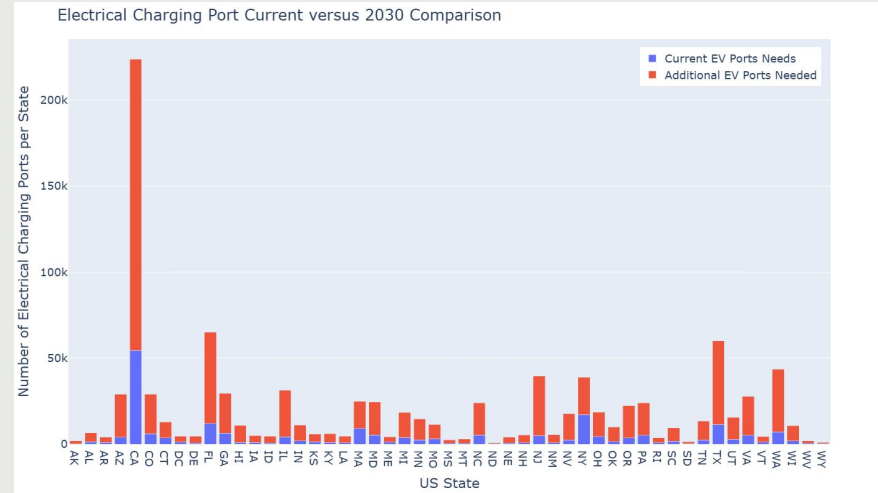
Results:

The power function predicts that **995,810 ports is needed to charge 29.7 million Battery EV's**, the forecasted amount of on-road electric vehicles by 2030 according to National Renewable Energy Laboratory (NREL) [3]. An additional 784,560 charging ports or 360,110 charging stations is to be built to our current infrastructure to meet 2030 demands..

Cost of Updating Infrastructure:

NREL estimates that 48% of charging ports will be public, 9% for Level 2 chargers and 39% for DC chargers [3]. The Alternative Fuels Data Center estimates an average equipment and installation costs for each type of charger:

- **Level 2 chargers - \$3,190 per charger**
- **DC chargers - \$104,000 per charge**



Projections to support 29.7 Mil Battery Electric Vehicle Growth by 2030:

+784.5 K Charging Ports



+360.1 K Charging Stations



\$66.8 B Infrastructure Investment



Future of Sustainable EV Infrastructure

Studies like the *MDPI 2024* analysis of consumer behavior in EV adoption suggest that:

“environmental consciousness and social pressure significantly influence individuals’ attitudes toward EVs, thereby enhancing their behavioral intentions to adopt such vehicles” [7].

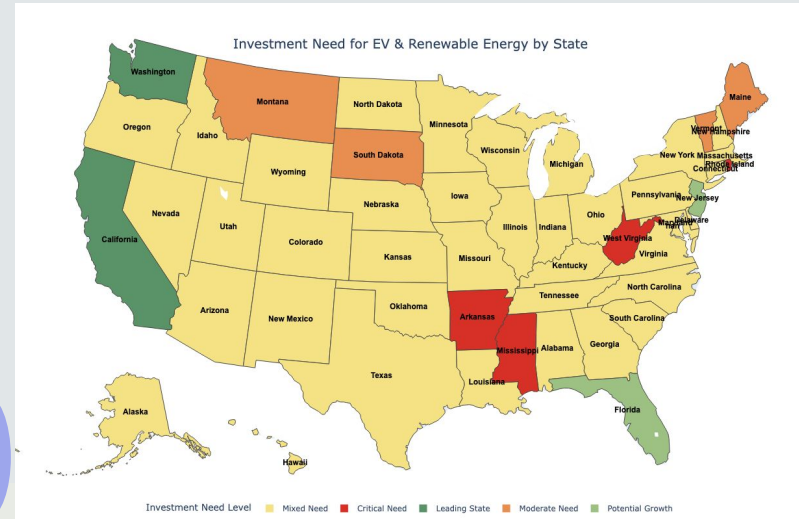
However, the environmental impact of EV adoption is largely determined by the energy sources powering the grid. As the U.S. Department of Energy’s Alternative Fuels Data Center notes:

“In geographic areas that use relatively low-polluting energy sources for electricity generation, all-electric vehicles and PHEVs typically have an especially large life cycle emissions advantage over similar conventional vehicles running on gasoline or diesel” [8].

One of the goals of this project is to identify outlier states with high EV adoption but low renewable energy usage. Electricity generation and emissions data from **eGRID** was used to examine the relationship between EV growth and grid sustainability. This analysis revealed significant regional disparities and identified several states that are in critical need of investment in EV infrastructure.

While states that have high clean energy like **California** and **Washington** empower their citizens to enjoy the full benefits of EV conversion; states like **Arkansas, Mississippi, and West Virginia** are in desperate need of infrastructure improvements. Despite trending EV adoption, these states remain fossil fuel dependent, limiting their potential emissions reductions.

Our earlier analysis of charging infrastructure investment trends adoption that current investments are not trending in the direction of states with the most need, further enforcing the need for targeted interventions like NEVI.



Project Summary and Next Steps

Q1: Are there currently enough charging stations within the U.S. to charge all electric vehicles?

Unfortunately, this question was not answered with our analysis. However, the relationship between EV population and EV infrastructure was observed. **As the EV population increases, the number of EV chargers increase exponentially by 0.77**

Q2: If there is a deficit, how many additional EV chargers per start are needed?

There currently isn't a deficit today, but there is expected to be a deficit by 2030. **29.7 million battery EV's** will be on the road [3]. To service all of the EV's, an additional **360.1 thousand stations or 784.5 thousand charge ports will need to be built to maintain current EV to EV port ratio.**

Q3: How much money is needed to increase the infrastructure to meet the EV demand?

To expand the current infrastructure to meet the 995,000 charge port demand by 2030, it is estimated to be a **66.8 billion dollar investment** for both equipment and labor.

Q4: In what specific states should infrastructure improvements be prioritized to not only increasing EV charging availability but also modernizing the energy grid to maximize the benefits of EV adoption?

If current trends were to persist, states with the most amount of EV's, **California, New York, Florida, and Texas**, should continue to expand their EV infrastructure due to higher demands. States that may not see as much of a carbon imprint improvement are states like **Arkansas, Mississippi, and West Virginia** and require modernization of the energy grid to see sustainability improvements.

Q5: Do some states see more benefits from EV adoption than others due to non-renewable resources?

Ideally, states with multiple renewable energy sources would be prioritized for infrastructure improvements over states that focus on non-renewable energy. The states with cleanest energy currently are **California and Washington.**

Next Steps

- **Confirm if the quantity of charging stations meet EV demand.** Validate if there is a golden ratio for electric vehicles to charging ports.
- **Analyze EV quantity per person per state to validate EV demand per state.** Review each state's population to confirm regional patterns.
- **Review which renewable resource can be utilized for each state toward EV infrastructure growth.** Ensure the most sustainable resources are analyzed by state.

Statements of Work and References

The whole team collectively organized the proposal and report to document our findings.

Jesse Garcia focused on the ethical and sustainable portion of the project. They researched on current impacts to US incentives to build EV infrastructure. They were able to apply their research and adapt it to their findings on sustainability and ease of implementation by state.

Janel Jackson created most data visualizations involving the current and future state of the infrastructure and EV population. In addition, she ran the correlation analysis to confirm the relationship between EV population and Quantity of Charge Ports.

Jason Rushton manipulated in cleaned all of the data set. In addition, they were able to confirm averages that were relevant to our analysis, but not necessarily calculated and were included in our assumptions. Jason also assisted with any troubleshooting with exporting the Jupyter Notebook file from Google Colab.

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[2] Davenport, C., & Ngo, M. (2025, February 7). *Trump Administration Move to Freeze E.V. Charger Funding Confounds States*. <https://www.nytimes.com/2025/02/07/climate/trump-ev-charger-funding-freeze.html>

[3] Wood, E., Borlaug, B., Moniot, M., Lee, D.-Y. (D-Y., Ge, Y., Yang, F., & Liu, Z. (2023, June). *The 2030 National Charging Network: Estimating U.S. Light-Duty Demand for Electric Vehicle Charging Infrastructure*. NREL Transforming Energy. <https://www.nrel.gov/docs/fy23osti/85654.pdf>

[4] U.S. Department of Energy (n.d.). *Electric Vehicle Charging Stations*. Alternative Fuels Data Center. <https://afdc.energy.gov/fuels/electricity-stations>

[5] Zargary, S. (2023, November 13). *How Much Electricity Does an Electric Car Use?* GenCellEnergy.com. <https://www.gencellenergy.com/resources/blog/ev-charging-power-car-electricity-usage/#:~:text=How%20Many%20Kwh%20Does%20an,1%20mile%20or%201.6km>

[6] St. John, J. (2022, October 25). 5 charts that shed new light on how people charge EVs at home. Canary Media. <https://www.canarymedia.com/articles/ev-charging/5-charts-that-shed-new-light-on-how-people-charge-evs-at-home>

[7] Rafiq, F., Parthiban, E. S., Rajkumari, Y., Nasir, M., & Dogra, N. (2023). *From Thinking Green to Riding Green: A Study on Influencing Factors in Electric Vehicle Adoption*. Sustainability, 16(1). <https://doi.org/https://doi.org/10.3390/su16010194>

[8] U.S. Department of Energy (n.d.). *Emissions from Electric Vehicles*. Alternative Fuels Data Center. <https://afdc.energy.gov/vehicles/electric-emissions>